Screening Test

№ 3568 **Junior 0991 ITMA** **J**

1. The quadrilateral formed by the angle bisectors of another quadrilateral is a

   a) rectangle h) rhombus c) square d) cyclic quadrilateral
   e) None of the above

2. Two circles with radii 2cms and 6cms have their centres at A and B where AB=10cms. PQ and RS are the transverse common tangents intersecting the line of centres at C. Then BC in cms is

   a) 2 b) 7.5 c) 8 d) 4 e) None of the above

3. A circle passes through the vertices of a triangle whose side lengths are 2.5, 6, 6.5. The radius of the circle is

   a) 6 5 b) 3.25 c) 2.5 d) 6 e) None of the above

4. If $yz : zx : xy = 1 : 2 : 3$ then $\frac{x}{yz} : \frac{y}{zx}$ is equal to

   a) 1 : 2 h) 2 : 1 c) 1 : 4 d) 4 : 1 e) None of the above

5. The number of real values of x for which the equation.
   $\left(2^{6x+3}\right)\left(4^{3x+6}\right) = 8^{4x+5}$ is satisfied is

   a) 2 b) 4 c) 8 d) infinitely many e) None of the above

6. If the ratio of $3x+4$ to $y+15$ is constant and $y = 5$ when $x = 2$, then, when $y=11$, x equals.

   a) .2 5 b) 2 c) 3 d) 2 . 5 e) None of the above

7. If two persons P and Q start at the same time from two points X and Y towards each other and after crossing they take x and y hours to reach Y and X respect.vely, then speed of P : speed of Q is

   a) $\sqrt{y} : \sqrt{x}$ b) y : x c) x : y d) $\sqrt{x} : \sqrt{y}$ e) None of the above

8. The altitudes of a triangle are in the ratio 3 : 4 : 5. Then the sides to which they are perpendicular are in the ratio

   a) 5 : 4 : 3 b) 3 : 5 : 4 c) 4 : 3 : 5 d) $\frac{1}{3} : \frac{1}{4} : \frac{1}{5}$ e) None of the above

9. The roots of the equation $abx^2 - (a^2 - b^2)x = ab$ are

a) $\frac{b}{a}$ and $\frac{a}{b}$ b) $\frac{b}{a}$ and $-\frac{b}{a}$ c) $-\frac{a}{b}$ and $\frac{b}{a}$ d) $-\frac{a}{b}$ and $-\frac{b}{a}$

e) None of the above

10. If $x = 0.01$, then the value of $\left(\log_{10} x\right)^{\log_{10} x}$ is

a) $\frac{1}{4}$ b) 4 c) $-\frac{1}{4}$ d) -4 e) None of the above

11. Let the product (12) (25) where each factor is written in base b be equal to 333 in base b. Then b equals

a) 5 b) 6 c) 7 d) 8 e) None of the above

12. The 3 digit number 4 a 3 is added to the 3 digit number 984 to give the 4 digit number 13 b 7 which is divisible by 11. Then the value of a+b is

a) 15 b) 11 c) 10 d) 12 e) None of the above

13. 4 lines parallel to the base of a triangle divide the other sides each into 5 equal segments and the area into 5 distinct parts. If the area of the largest part is 27, then the area of the original triangle is

a) 54 b) 75 c) 81 d) 100 e) None of the above

14. Let F = 0.48181... be an infinite repeating decimal with the digits 8 and 1 repeating. When F is written as a fraction in lowest terms then the denominator exceeds the numerator by

a) 13 b) 14 c) 29 d) 57 e) None of the above

15. The number of numbers which when divided by 4 leave a remainder 2, when divided by 5 leave a remainder 3, when divided by 6 leave a remainder 4 and when divided by 7 leave a remainder 5 is

a) 0 b) 1 c) 2 d) Infinite e) None of the above

16. Let ABC and DEF be 2 triangles such that AB = DE, AC = DF and the median AX = median DY. Then

a) BC = EF b) BC < EF c) BC > EF

d) BC cannot be determined e) None of the above

17. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ where is $\mathbb{R}$ the set of real numbers be defined as $F[x] = [x + \frac{1}{2}]$ where $[y]$ is the greatest integer $\leqslant y$. Then $F \circ F$ is

a) F b] $-F$ c) a Constant function d) identity function
e] None of the above

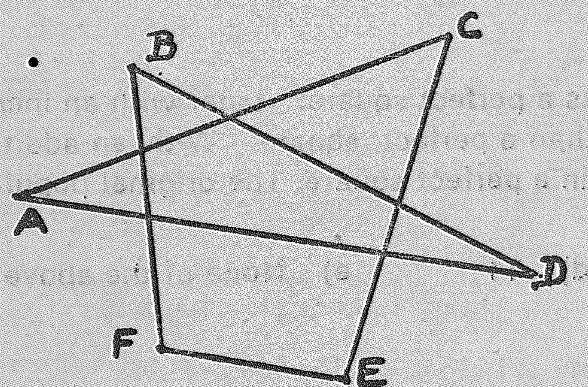
18. Point F is taken on side AD of a square ABCD. At C a perpendicular is drawn to CF meeting AB extended in E. The area of ABCD is 256 square cms and the area of $\triangle$ CEF is 200 square cms. Then the length of BE is

a) 12 cms b) 14 cms c) 15 cms d) 16 cms
e] None of the above

19. Let $S = \{1, 2 \ldots\ldots 80\}$. Define $a \star b$ = remainder of ab when divided by 81. The number of ordered pairs (a, b), $a, b \in S$ such that $a \star b \notin S$ is

a) 4 b) 2 c) 3 d) 5 e) None of the above

20.



In the above figure if $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 90n^\circ$, then n is equal to

a) 2 b) 3 c) 4 d) 5 e) None of the above

21. In a triangle the area is numerically equal to the perimeter. The radius of the inscribed circle is

a) 3 b) 1 c) 1.5 d) 2 e) None of the above

22. The greatest integer that will divide 13511, 13903, 14589 and leave the same remainder is

a) 28 b) 49 c) 98 d) 196 e) None of the above

23. Given $n! = 1.2.3 \ldots n$, let $S = 1! + 2! + 3! + 4! + \ldots + 1000!$ the units digit in S is

a) 1 b) 0 c) 4 d) 3 e) None of the above

24. The number of integers for which $x^2 + 6x + 12$ is a perfect square is

a) 0 b) 1 c) 2 d) 3 e) None of the above

25. The number of digits in $2^{12} \times 5^8$ is

a) 9 b) 10 c) 11 d) 12 e) None of the above

26 The number of solutions of $\{1, 2\} \subseteq X \subseteq \{1, 2, 3, 4, 5\}$ where X is a set is

a) 2 b) 4 c) 6 d) 8 e) None of the above

27. The population of Madras at one time was a perfect square. Later with an increase of 100, the population was one more than a perfect square. With an additional increase of 100, the population was again a perfect square. The original population is a multiple of

a) 3 b) 7 c) 9 d) 11 e) None of the above

28. The last digit of the number 3^{400} is

a) 0 b) 2 c) 1 d) 3 e) None of the above

29. If $= y \sqrt{2 + \sqrt{2 + \sqrt{2 \ldots}}}$ then y is

a) 8 b) 4 c) 3.14 d) 2 e) None of the above

30. $123456^{654321} + 3456789^{9876543}$ when divided by 5 leaves the remainder

a) 0 b) 1 c) 2 d) 3 e) None of the above